Data Science & AI Engineer | Generative AI

Benaboudi Saad

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Profile

Final-year Computer Engineering student (graduating 2027) with hands-on experience in **data preparation, machine learning and deep learning models, and generative AI** (RAG, LLM integration), exposed through **REST APIs** and deployed with **Docker**. Seeking an **end-of-studies internship (PFE) in Data Science / AI** with the prospect of a full-time offer.

Experience

AI Software Engineer July – September 2026

SEGULA Technologies *Casablanca, Morocco*

- Designed and built a **multimodal generative AI pipeline (RAG)** (Python, FastAPI, LangChain, ChromaDB, PostgreSQL, MinIO) for extraction, summarization and semantic search over vehicle homologation technical documents.
- Optimized vector search relevance to reduce the noise passed to the LLM and improve the quality of generated answers.
- Developed and exposed **APIs** to serve AI features (JWT authentication, role-based access) and implemented **resilience** mechanisms (retries, fallback between LLM providers) to handle rate limits of external services.
- Designed a **100% local** target architecture based on open-source LLMs to meet the data confidentiality requirements of industrial data.
- Containerized the whole application with **Docker**

Full-Stack Engineer August 2025

Royal Air Maroc – Digital Factory *Casablanca, Morocco*

- Designed and developed an employee attendance management system.
- Built a REST backend with Spring Boot and PostgreSQL.
- Developed the frontend interface with React and TypeScript.
- Implemented JWT authentication and role management (admin / employee).
- Took part in the full development cycle: requirements analysis, design and implementation.

Academic Projects

Employee Attrition and Salary Prediction | *Python, Scikit-learn, Pandas, Matplotlib, Seaborn*

- **Data preparation**: (EDA) and preprocessing of the IBM HR Analytics dataset (data cleaning, categorical encoding, feature engineering).
- Trained and **evaluated** classification models (attrition) and regression models (salary),identified key factors through correlation analysis and visualizations.

Intelligent Crop Analysis | *Python, TensorFlow, CNN, Gradio*

- Built and trained a **deep learning** image recognition model (CNN) with TensorFlow, and made it usable through a real-time demo interface (Gradio).

KnowBase — Personal RAG Platform | *Spring Boot, Python, FastAPI, LangChain, pgvector, Docker*

- Microservices architecture combining a Spring Boot backend (REST API, JWT), a Python generative AI service (RAG) and an Android mobile app, deployed with Docker Compose.

Education

National School of Applied Sciences (ENSA) Berrechid, Morocco

Engineering Degree – Computer Engineering *Sep. 2022 – June 2027*

Technical Skills

Languages: Python, SQL, Java, TypeScript, Bash
Data Science & ML: Scikit-learn, TensorFlow, CNN, pandas, NumPy, Matplotlib, Seaborn, EDA
Generative AI: LLMs, RAG, LangChain, embeddings, semantic search, prompt engineering, ChromaDB, pgvector
APIs & Backend: FastAPI, Spring Boot, Node.js, REST, JWT, Spring Security, JPA/Hibernate
DevOps & Tools: Docker, Docker Compose, Git, Linux, PostgreSQL, MySQL, MinIO